Fernando VAN DER VINNE

Mechanical Design Engineer | Robotics & Product Development

linkedin.com/in/fernando-vinne   Munich, Germany
Portfolio
EU citizen

Mechanical Design Engineer with hands-on experience in robotics, product development, CAD design, prototyping, and mechanical systems integration. Strong background in developing robot components, end-effectors, 3D-printed parts, and functional prototypes from concept to testing. Passionate about creating practical engineering solutions at the intersection of mechanical design, automation, and robotics.

PROFESSIONAL EXPERIENCE & ENGINEERING PROJECTS

Present – Sep 2024

Mechanical Design Engineer, sewts GmbH, Munich
Full-time Intern: Sep 2024–Sep 2025 | *Working Student: Sep 2025–Present*

- Served as primary mechanical designer for key hardware modules of a robotic cell for automated textile handling in industrial laundries
- Developed a functional textile gripper concept that addressed a long-standing technical challenge in automated laundry handling and was used in 2 deployed gripper prototypes
- Designed and tested 5+ gripper concepts and 20+ rapid-prototyped or silicone-cast iterations to improve textile pickup performance under realistic laundry conditions
- Contributed to 2 deployed versions of a dirty-laundry separator system, from early concept to industrial deployment, with a 3rd version currently in development
- Designed robot end-effectors, silicone fingers, gripper structures, pneumatic concepts, robot cabling, fixtures, transport and assembly concepts for industrial use
- Developed CAD assemblies, molds, 3D-printed parts, mechanical test setups, and hardware used for AI training data collection in customer laundry environments

Autodesk Inventor | End-Effectors | Gripper Design | Pneumatics | Silicone Casting | Prototyping | Industrial Automation

Aug 2024 – Sep 2021

Research Assistant - Mechanical Design & Robotics, TUM LPL, Garching

- Designed and developed mechanical components for robotic systems, including joints, structural parts, and test assemblies
- Created and optimized CAD models for research prototypes, manufacturing, assembly, and finite element analysis
- Supported the development of robotic arm modules, with a focus on mechanical stability, load transfer, bearing integration, and modularity
- Managed and maintained FDM and SLA 3D printers used for prototyping and research projects

Fusion 360 | Product development | FDM and SLA 3D Printing

May 2024 – Dec 2023

Bachelor Thesis – Modular Low-Cost Robot Arm, Grade: 1.3, TUM LPL, Garching

- Developed modular mechanical concepts for low-cost robotic arm systems, improving stability, load handling, and assembly accessibility
- Redesigned key robot arm modules through improved bearing selection, load distribution, and mechanical integration
- Addressed alignment, compatibility, and maintenance challenges in a modular robotic system
- Designed end-effectors for specific laboratory automation tasks

Robot Arm Design | Modular Design | Bearing selection | Mechanical Integration | End-Effectors

Feb 2023 – Oct 2022

Project Seminar - Exoskeleton Knee Joint, TUM LPL, Garching

- Developed a knee joint concept for an exoskeleton supporting patients with limited movement capacity
- Contributed to the design of a series elastic actuator, combining mechanical design, electronics, control, simulation, and prototyping

Exoskeleton | Series Elastic Actuator | Mechatronics | Prototyping | Product Development

Sep 2019 – Aug 2019 | **Manufacturing Intern, Reindl Maschinenbau GmbH, Glonn**

- Gained hands-on experience in manufacturing, assembly, quality control, and CNC-based production processes
- Supported the production and installation of mechanical equipment, mainly for automotive manufacturing applications

Manufacturing | Assembly | Quality Control | CNC | Automotive Production

Education

Dec 2027	**M.Sc. Mechanical Engineering** - Technical University of Munich *(expected)*
2019–2025	**B.Sc. Mechanical Engineering** - Technical University of Munich
2018–2019	**Studienkolleg T-Kurs** - TU Dresden, Institute of Advanced Studies

Technical Skills

CAD & Mechanical Design	Autodesk Inventor & Vault, Fusion 360, SolidWorks, CATIA V5, complex assemblies, technical drawings, design for manufacturing
Robotics & Automation	Robot end-effectors, gripper design, robot fixtures, pneumatic systems, cable routing, sensor/mechanical integration
Prototyping & Manufacturing	FDM/SLA 3D printing, silicone casting, mold design, mechanical testing, assembly, CNC/manufacturing basics
Simulation & Engineering Tools	MATLAB/Simulink, finite element analysis basics, Python, C/C++
Product Development	Concept development, iterative prototyping, mechanical validation, design reviews, practical problem solving

Languages

- **Portuguese** – Native
- **English** – C1, TOEFL certificate
- **German** – C1, Studienkolleg and B.Sc. Mechanical Engineering at TUM completed in German
- **Spanish** – B1, TUM language course